BioNutrient Supply Chain Optimization

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Just a BI student

I recommend viewing this as an html in Your webbrowser for the best experience. PDF is a stillshot of the html.

Assumptions: Since no initial stock is provided I assume it is 0.
The official AMPL style guide will be used.I will add an underscore to constraints to make them easier to differentiate.

Sets

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Parameters

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Decision Variables

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Constraints

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Objective Function

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Data

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```
HiGHS 1.11.0HiGHS 1.11.0: optimal solution; objective 1006501083
503299 simplex iterations
3323 branching nodes
absmipgap=93422.4, relmipgap=9.28189e-05
```

Checking values

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Answers

Question 1: Should the company rent the central warehouse?

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```
Answer 1: Yes, the company should rent the central warehouse (Cost: NOK 7,000,000).
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Question 2: How many shifts should be run at each plant per quarter?

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```
Answer 2: Quarterly Shift Plan (Min 3, Max 5 shifts):
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-------------
        Q1  Q2  Q3  Q4
index0
North    3   4   5   3
South    3   3   5   3
```

Question 3: How much should be produced at each plant per quarter?

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```
Answer 3: Quarterly Production Volume (Tons):
-------------------------------------------------------------------------------------
-------------
             Q1       Q2       Q3       Q4
index0
North   51000.0  68000.0  85000.0  51000.0
South   10511.0  45000.0  73489.0  45000.0
```

Question 4: How much should be transported between warehouses?

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```
Answer 4: Full List of Inter-Warehouse Transfers (Tons):
------------------------------------------------------------------------------------------
-------------
                           WarehouseTransfer.val
Boat    Route Leg Quarter
Anna    R9    1   2                 20475.000000
              2   2                  4220.833333
Belinda R9    1   1                 19125.000000
Diana   R2    1   4                 10387.500000
Gunda   R2    1   1                  4500.000000
                  3                  7666.666667
```

Question 5: How much should be transported from each warehouse to each sales district in each quarter?

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```
Answer 5: Quarterly Shipments to Sales Districts (Tons):
------------------------------------------------------------------------------------------
-------------
Q                1        2        3        4
W       D
Central D1   739.0      0.0      0.0      0.0
        D2     0.0      0.0      0.0      0.0
        D3     0.0   7962.0  11917.0      0.0
        D4     0.0      0.0   7667.0  10388.0
        D5  7875.0   7962.0   3423.0      0.0
North   D1  3261.0  19000.0  34000.0  15000.0
        D2  3000.0  15000.0  21000.0  14000.0
        D3  7000.0   9746.0  21083.0  13033.0
        D4  6614.0  11108.0     -0.0      0.0
        D5     0.0      0.0      0.0      0.0
South   D1     0.0      0.0      0.0      0.0
        D2     0.0      0.0      0.0      0.0
        D3     0.0   6292.0      0.0  11967.0
        D4  3386.0  12892.0  28333.0   6613.0
        D5  7125.0  30038.0  44577.0  27000.0
```

Question 6: How much should be stored at each of the warehouses by the end of each quarter?

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```
Answer 6: End-of-Quarter Storage Levels (Tons):
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```

	Initial	Q1	Q2	Q3	Q4
index0					
Central	0.0	15011.0	15340.0	0.0	0.0
North	0.0	7500.0	171.0	1421.0	0.0
South	0.0	0.0	0.0	579.0	0.0

Question 7: Which boats should be used in the transportation?

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```
Answer 7: Boat Hiring Decisions
-------------------------------------------------------------------------------------
-------------
- Belinda: Full Year Rental
- Diana: Full Year Rental
- Emma: Full Year Rental
- Gunda: Full Year Rental
- Anna: Q2+Q3 Rental Only
```

Question 8: Which route should each boat follow in each quarter?

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```
Answer 8: Boat-to-Route Assignments per Quarter
-------------------------------------------------------------------------------------
-------------

--- Q1 ---
  - Belinda  is following Route R9
  - Diana    is following Route R1
  - Emma     is following Route R8
  - Gunda    is following Route R2

--- Q2 ---
  - Anna     is following Route R9
  - Belinda  is following Route R8
  - Diana    is following Route R1
  - Emma     is following Route R3
  - Gunda    is following Route R7

--- Q3 ---
  - Anna     is following Route R3
  - Belinda  is following Route R8
  - Diana    is following Route R1
  - Emma     is following Route R7
  - Gunda    is following Route R2
```

```
--- Q4 ---
  - Belinda  is following Route R8
  - Diana    is following Route R2
  - Emma     is following Route R3
  - Gunda    is following Route R7
```

Question 9: How much should be supplied by each boat to each sales district and to each warehouse each quarter?

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```
Answer 9: Detailed Supply per Boat (Tons)
-----------------------------------------------------------------------------------------
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```

	Quarter	1	2	3	4
Boat	**Destination**				
Anna	Central	0.0	20475.0	0.0	0.0
	D1	0.0	0.0	34000.0	0.0
	D2	0.0	0.0	21000.0	0.0
	South	0.0	4221.0	0.0	0.0
Belinda	Central	19125.0	0.0	0.0	0.0
	D3	0.0	1783.0	13033.0	25000.0
	D4	0.0	24000.0	13033.0	1067.0
Diana	Central	0.0	0.0	0.0	10388.0
	D1	4000.0	0.0	0.0	0.0
	D3	4614.0	22217.0	19967.0	0.0
	D4	0.0	0.0	7633.0	15933.0
	D5	15000.0	7962.0	3840.0	0.0
Emma	D1	0.0	19000.0	0.0	15000.0
	D2	0.0	15000.0	0.0	14000.0
	D3	2386.0	0.0	0.0	0.0
	D4	6614.0	0.0	0.0	0.0
	D5	0.0	0.0	44160.0	0.0
Gunda	Central	4500.0	0.0	7667.0	0.0
	D2	3000.0	0.0	0.0	0.0
	D4	3386.0	0.0	15333.0	0.0
	D5	0.0	30038.0	0.0	27000.0

Question 10: What will be the total yearly costs for your plan? Specify the individual cost components (raw material costs, production capacity costs, etc...)

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Master Cost Breakdown
--------------------------------------------------------------------------------------------
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Category                        Cost (NOK)
--------------------------------------------------------------------------------------------
-------------
Raw Material Costs:            807,000,000
Fixed Production:              116,000,000
Overtime Costs:                 10,500,000
Storage Holding:                 2,001,083
Warehouse Rent:                  7,000,000
Boat Rental:                    64,000,000
Shortfall Penalties:                     0
--------------------------------------------------------------------------------------------
-------------
TOTAL YEARLY COST:           1,006,501,083
```

Question 11: Imagine that your plan is rejected because it does not consider the uncertainty that will be present in a practical situation and its impact on customer service. What types of uncertainty could that be? How would you deal with this, that is, how would you take into account uncertainty and customer service?

Types of Uncertainty

A deterministic optimization model assumes all input parameters are fixed and known. In a real-world maritime supply chain, several variables introduce uncertainty that can lead to significant cost variances:

1. Demand Volatility: Customer demand in the five districts is rarely static. Weather conditions or market shifts can cause spikes in Q2 or Q3. If demand exceeds the forecast even slightly, the model may incur massive Shortfall Penalties (12,000 NOK/ton) because the fleet capacity is already optimized to the limit.

2. Operational and Lead-Time Risks: The model assumes fixed travel times (e.g., 16 days for Route 1). In reality, North Sea storms, port congestion, or mechanical breakdowns can delay a boat. A 10-day delay for a vessel like Anna during the peak season could result in empty shelves at a sales district.

3. Production Uncertainty: Mechanical failures at the North or South plants could reduce the weekly output. Since the current plan minimizes inventory to save on holding costs, there is

no “buffer” to protect against a production halt.

Dealing with Uncertainty and Customer Service

To address these rejections, I would move from a “Minimum Cost” strategy to a “Robustness” strategy. This involves the following steps:

1. Building Physical Buffers (Safety Stock) The current model often targets zero inventory at the end of periods to save on the 50 NOK/ton holding cost. To prioritize customer service, I would add a Safety Stock Constraint, requiring each warehouse to maintain a minimum inventory level (e.g., 15% of the next quarter’s expected demand). This acts as an insurance policy against lead-time delays or demand spikes.

2. Increasing the Penalty Weight (Service Level Focus) If customer service is the priority, the “Shortfall Penalty” should be set high enough to ensure the solver treats it as a “hard” constraint. By increasing the penalty significantly above the cost of hiring an additional boat (like Carla), the model is forced to hire extra capacity “just in case” rather than risking a stock-out.

3. Implementing Stochastic Programming Instead of solving for one set of numbers, I would use Scenario-Based Optimization. This involves testing the plan against “Best Case,” “Worst Case,” and “Most Likely” demand scenarios simultaneously. This allows the model to identify “first-stage” decisions (like which boats to hire for the full year) that perform well across all possible futures, rather than a plan that only works if everything goes perfectly.

4. The Information Trade-off I would argue that better data and predictive analytics can substitute for expensive physical assets. By improving demand forecasting, we can “pre-position” inventory in the Central warehouse during the quieter Q1/Q2 periods. This uses the cheaper storage capacity to mitigate the risk of needing an expensive emergency boat hire during the Q3 peak.

Summary:

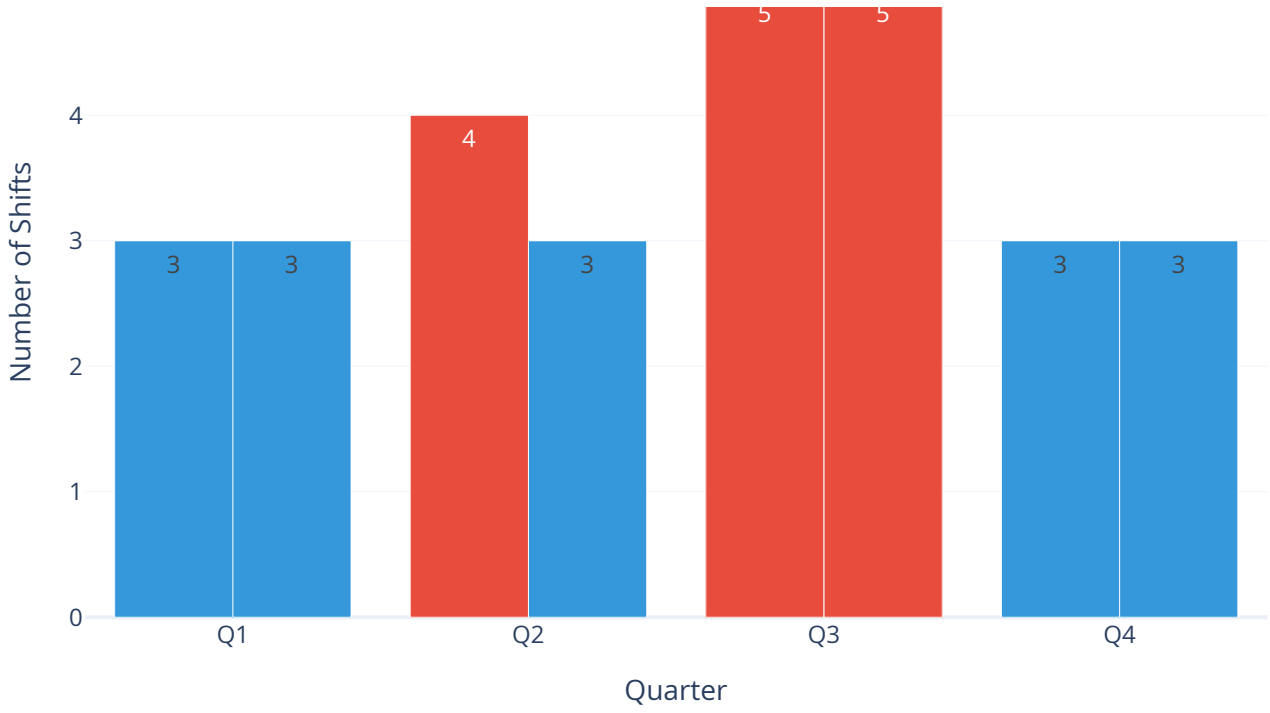
Uncertainty creates a Cost Variance. A plan that is “optimal” on paper may become the most expensive option if it is too fragile to handle a storm or a demand spike. By incorporating safety stocks and scenario testing, we ensure that the total yearly cost remains stable and customer service is guaranteed, even when reality deviates from the forecast.

Visualization

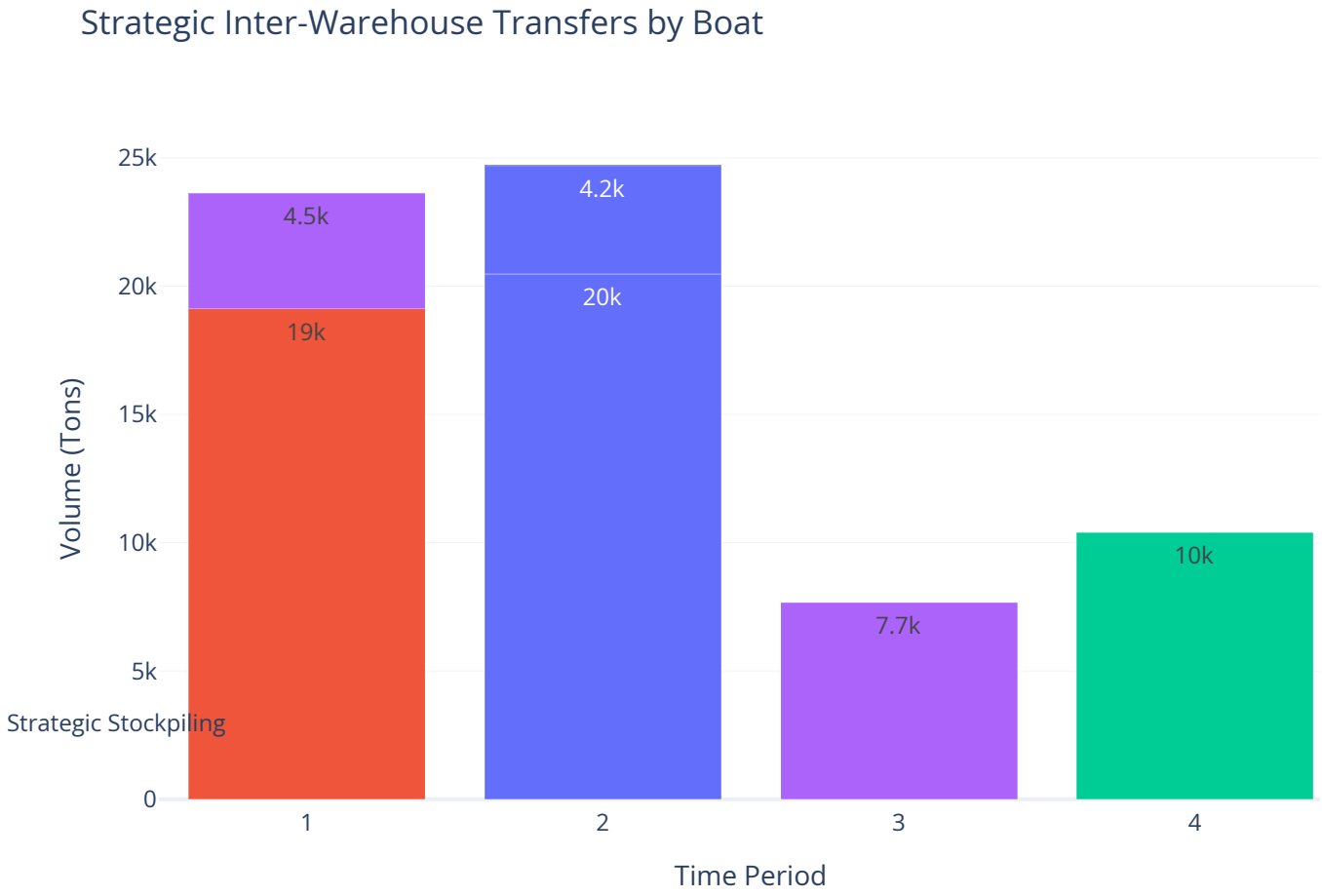
This is a section I am making as a bonus to show the coolness of this submission format

▶ Click to expand the plot code

Quarterly Production Shifts: Overtime Detection

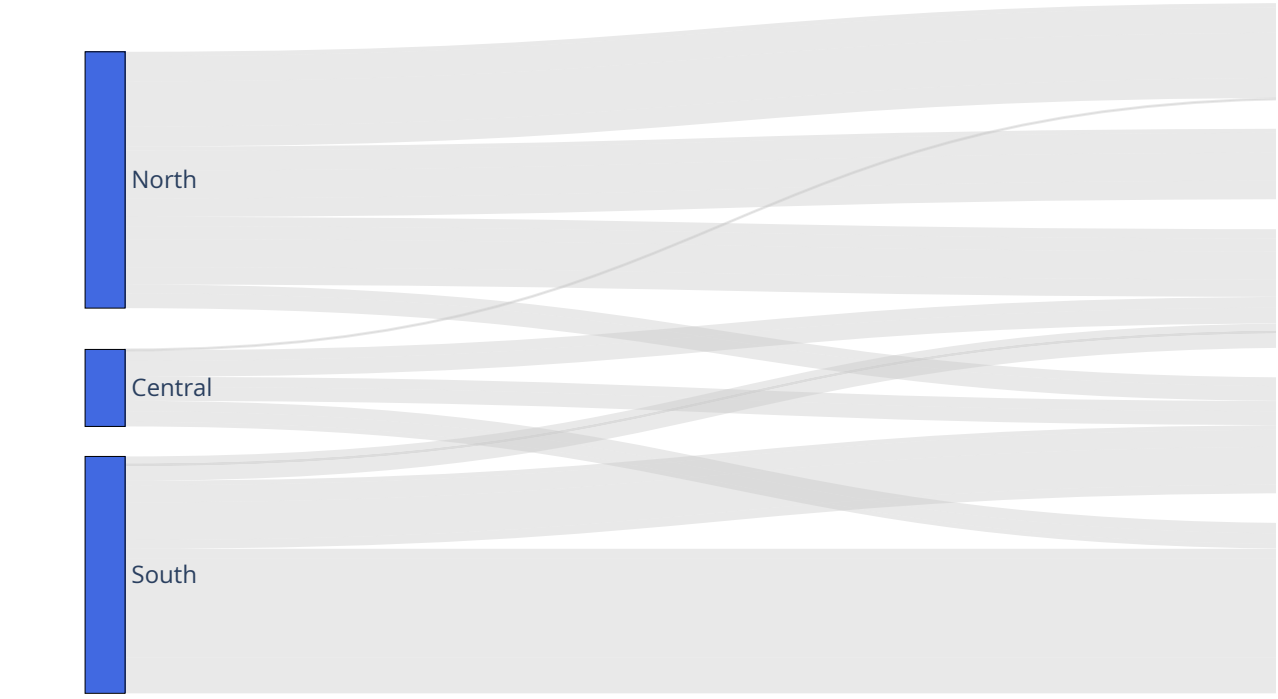


▶ Click to see Plotly code for Warehouse Transfers

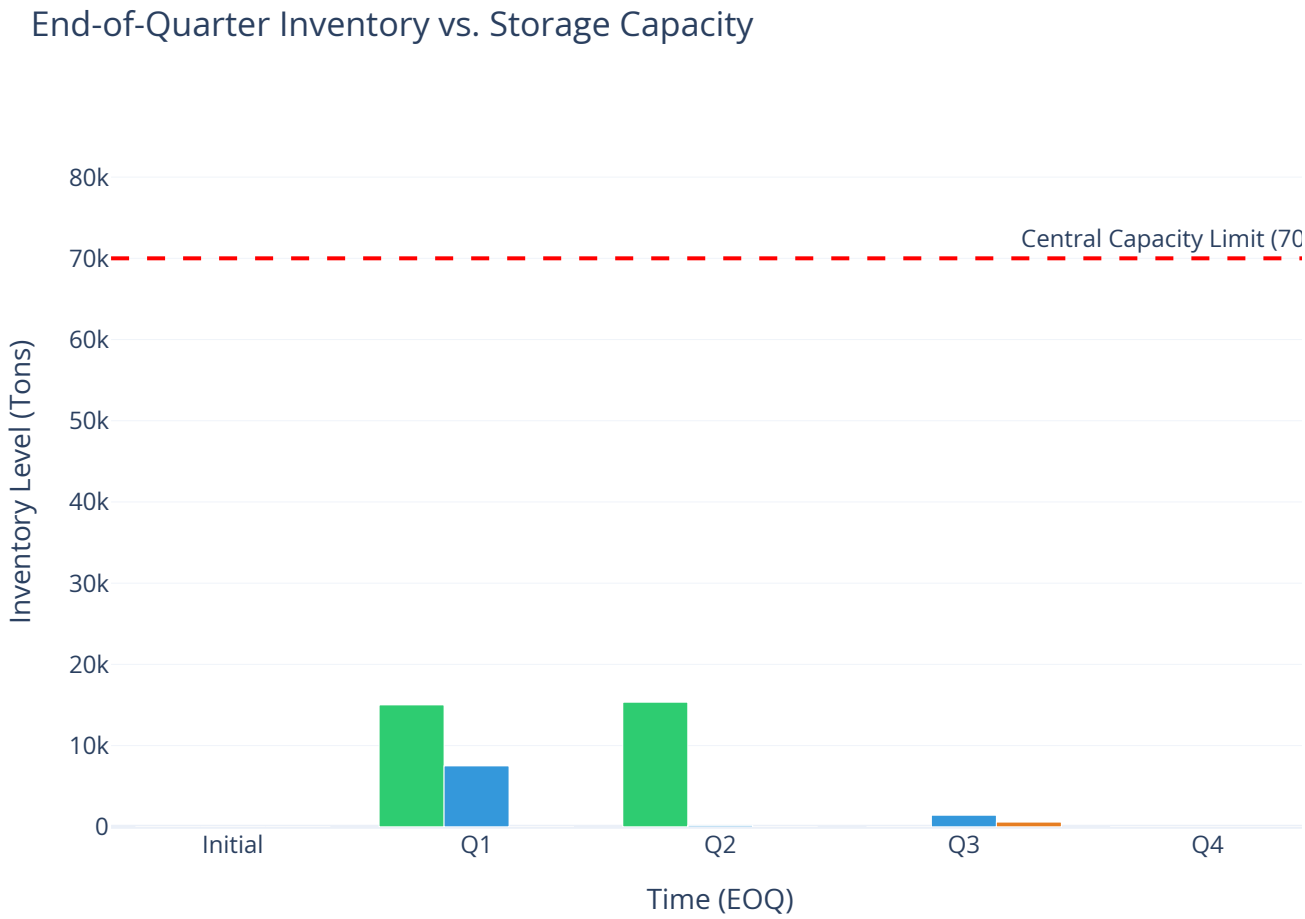


▶ Click to see Sankey Flow Diagram code

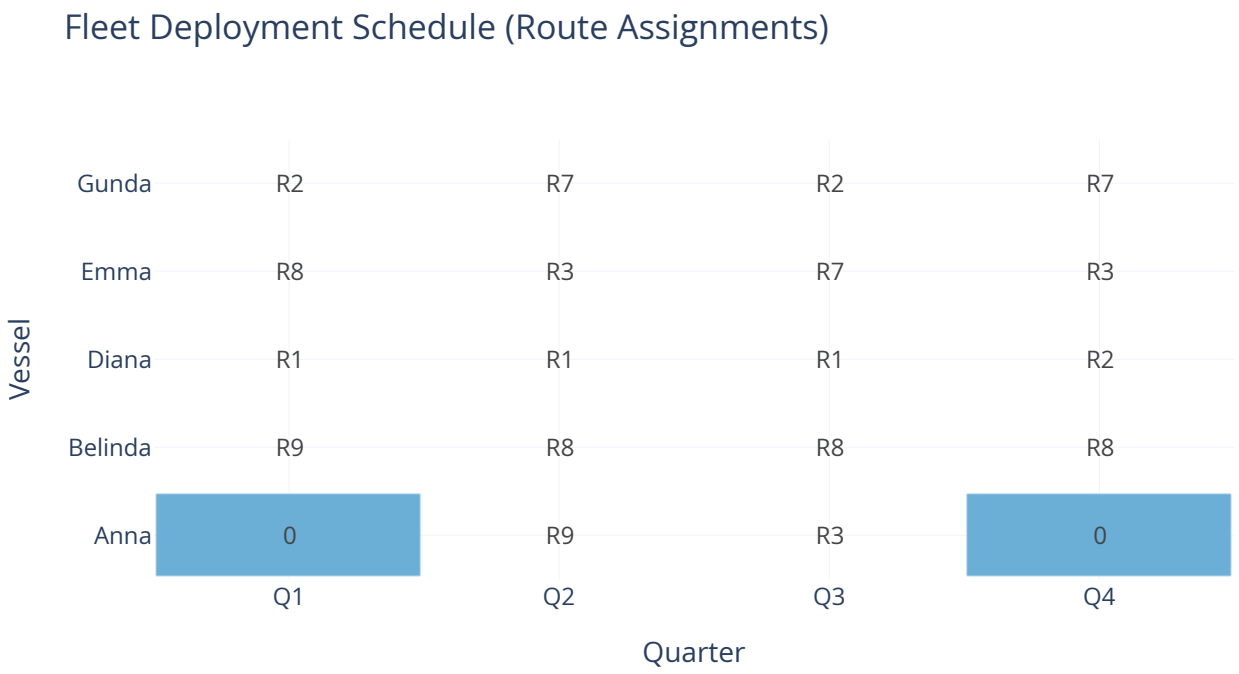
Supply Chain Flow: Origin to Districts (Sankey Diagram)



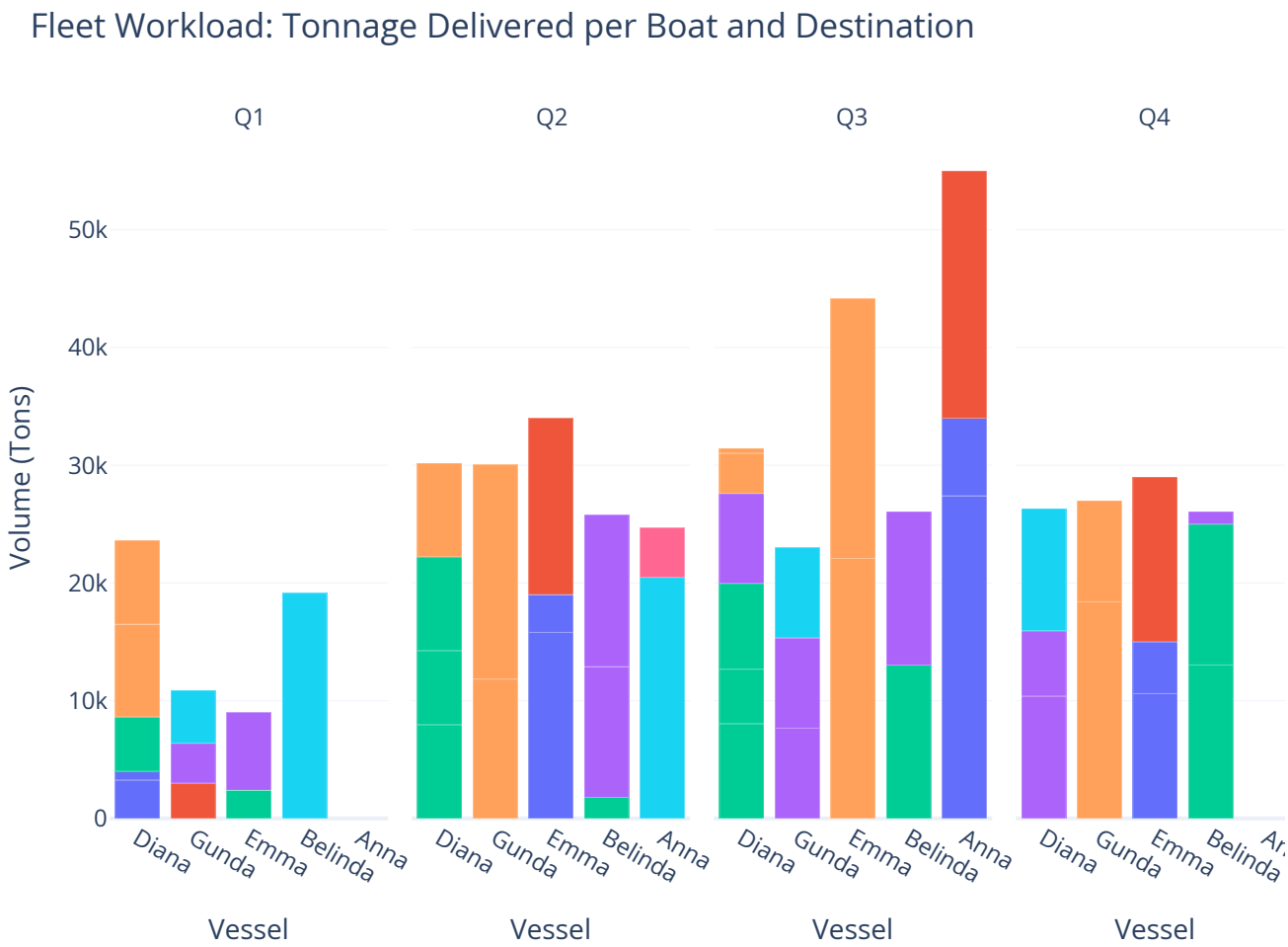
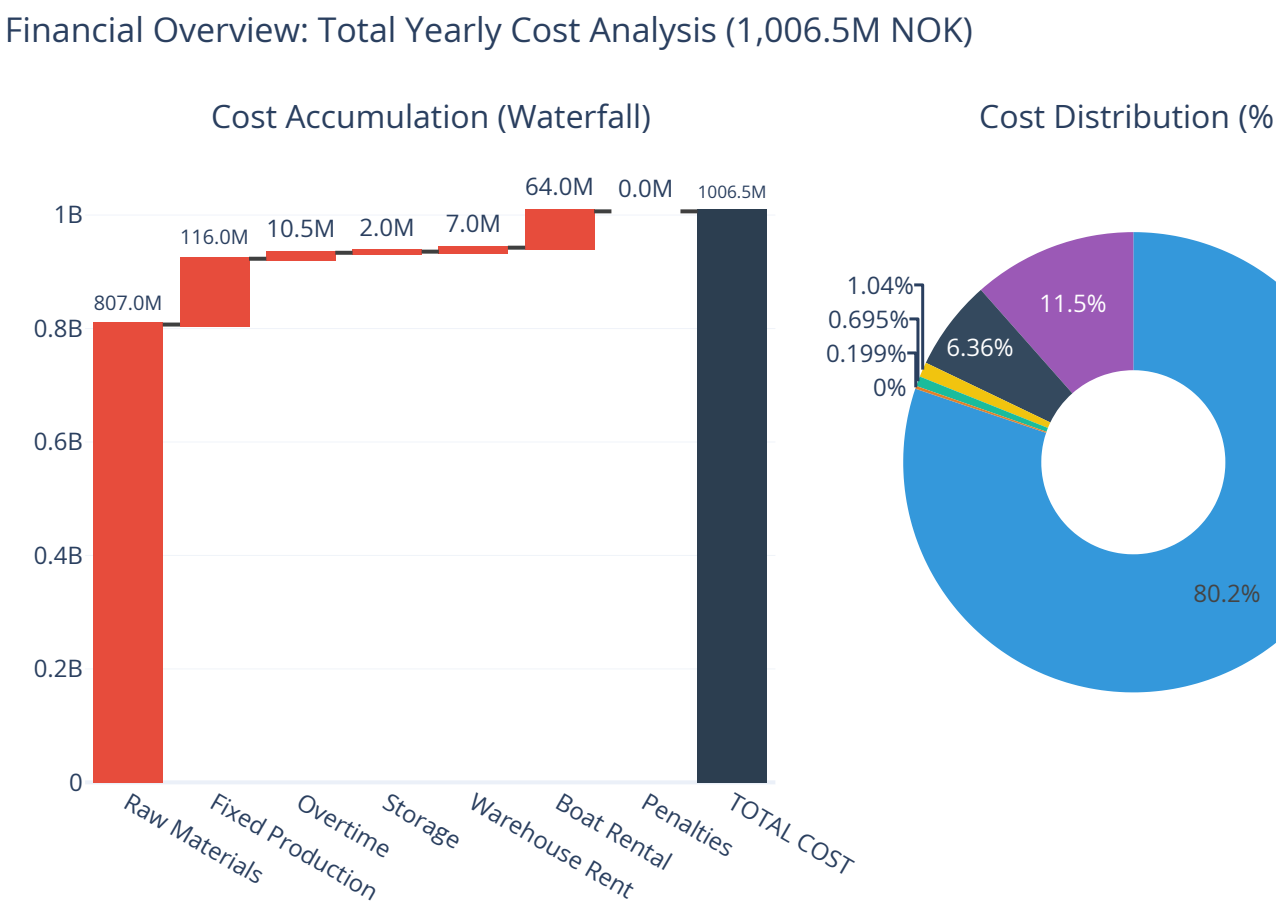
▶ Click to see Plotly code for Warehouse Capacity Utilization



▶ Click to see Plotly code for Boat-to-Route Assignments



▶ Click to see Boat-to-Destination Volume Plot



▶ Click to see Plotly code for Cost Breakdown